Total No. of Questions : 8]

SEAT No. :

PE-2487

[Total No. of Pages : 3

[6583]-11
T.E. (Civil)
WATER SUPPLY ENGINEERING
(2019 Pattern) (Semester - V) (301002)

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates:

1) ***Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, and Q.7 or Q.8***
2) ***Figures to the right indicates full marks.***
3) ***Draw neat figures wherever necessary.***
4) ***Assume necessary data.***
5) ***Use of scientific calculator is allowed.***

Q1) a) State the principle of coagulation and flocculation. Differentiate between coagulation and flocculation. **[8]**

b) A filter unit has surface area of 5×10 m^2. After filtering 10,000 m^3 for 40 h, the filter is backwashed for 15 min at a rate of 0.5 m/min. Find: (i) the average filtration rate, (ii) the quantity of wash water, (iii) percent of wash water to treated water, and (iv) the flow rate to each of the four troughs. **[6]**

c) Describe the operational troubles in filter. **[4]**

OR

Q2) a) A rapid mixer is to be used for coagulation of surface water with high turbidity. If the flow is 650 m^3/h, find the volume and dimensions of the square tank and the power requirements. Assume that the detention time is 20 seconds and G=1000 s^{-1} and μ = 0.00115 N s/m^2 at T=15°C **[8]**

b) Mention various factors affecting coagulation. **[6]**

c) Discuss the remedial measures for various operational troubles in filter. **[4]**

P.T.O.

Q3) a) A water treatment plant uses 65 kg/d of chlorine to treat 50 MLD of water. The residual chlorine after 30 min contact time is 0.15 mg/L. Determine the chlorine dosage and chlorine demand of the water. **[7]**

b) Explain fluoridation and defluoridation. **[6]**

c) Mention any two different disinfectants and state their characteristics. **[4]**

OR

Q4) a) Enlist and explain various methods of disinfection. Explain the factors affecting the efficiency of disinfection. **[7]**

b) The water works of a town of population 100000 has to meet its water demand at the rate of 135 lit/capita/day. If the disinfection is to be done by bleaching powder having 40% available chlorine, determine the quantity of bleaching powder required per year. The required dose of chlorine at the water works is 0.3 ppm for disinfection. **[6]**

c) State the principle and advantages of water softening by ion exchange method. **[4]**

Q5) a) Describe any four methods for water distribution in detail. **[8]**

b) Explain detection and prevention of leakage. **[6]**

c) Explain balancing capacity of reservoir. **[4]**

OR

Q6) a) Determine the balancing capacity of an elevated service reservoir for a town having a population of 2 million and water supply rate of 280 l/c/d. The water is pumped continuously for 24 hrs. The breakup of demand is as follows. **[8]**

Time	3am-9am	9am-1pm	1pm-7pm	7pm-11pm	11pm-3am
Demand in l/c/d	80	50	85	30	35

b) Mention requirements for good distribution system. **[6]**

c) Write the necessity of rainwater harvesting system. **[4]**

Q7) a) Write and explain use of any three fixtures and any three fittings used for water supply system. **[7]**

b) Explain SMART city mission and AMRUT mission for improvement of infrastructure sector. **[6]**

c) List and draw any four sanitary pipe fittings. **[4]**

OR

Q8) a) Discuss the concept of packaged water treatment plants for townships and mention advantages. **[7]**

b) Explain Jal Jeevan Mission and its impact in rural India before and after its implementation. **[6]**

c) Discuss WTP for swimming pools. **[4]**